

Amit Kumar Mahato

Backend Developer (Scalable Systems & Performance)

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SUMMARY

Backend-focused developer with experience building scalable, secure systems and optimizing high-concurrency APIs. Strong in multi-tenant architecture, system design, performance optimization, and backend reliability.

PROJECTS

Manage In Sync

Jan 2026 – Present

Live: manageinsync-frontend.onrender.com

GitHub: github.com/Amit-Mahato-0001/ManageInSync

- Designed and built a multi-tenant SaaS platform enabling agencies to manage dashboards, projects, clients, members, and tasks with strict tenant-level data isolation
- Architected tenant-scoped RBAC (owner/admin/member/client), invite-based onboarding, protected APIs, validation, and audit tracking ensuring secure access and zero cross-tenant leakage
- Implemented authentication using short-lived JWTs, rotating refresh tokens, and session tracking to prevent unauthorized access and session abuse
- Strengthened backend security with CORS allowlists, rate limiting, CSP headers, secure cookies, and environment-based secret management
- Delivered collaboration and billing workflows including conversations, invoice generation, payment tracking, and Razorpay verification
- Built containerized deployment using Docker, Docker Compose, and CI/CD pipelines with GitHub Actions for automated testing, build, and deployment
- Optimized performance using indexing, pagination, filtering, lean queries, and load testing with consistent low latency and zero-error responses under concurrency

Backend Performance Lab

Feb 2026 – Present

GitHub: github.com/Amit-Mahato-0001/Scaling-API

- Designed and executed load testing experiments to evaluate API performance under varying latency and concurrency using Autocannon
- Improved throughput from 9 req/sec to 5000 req/sec by eliminating I/O bottlenecks and optimizing request handling
- Implemented Redis caching and rate limiting to reduce latency and improve system stability under load
- Scaled throughput to 9700 req/sec using Node.js clustering and multi-core utilization
- Identified Redis rate limiting overhead as a bottleneck through controlled benchmarking
- Achieved peak throughput of 14000 req/sec at optimal concurrency while maintaining low latency

Pixelva

Jan 2026 – Present

Live: [Edge Add-on Store](#)

GitHub: github.com/Amit-Mahato-0001/Pixelva

- Built a Chrome extension for screenshot capture and editing with a browser-based UI
- Designed a canvas-based rendering system with support for text, shapes, and resizing
- Implemented background service workers and message passing for stable performance

SKILLS

Backend & APIs: Node.js, Express.js, REST APIs, JWT, API Security

Databases: MongoDB, Mongoose, Indexing, Query Optimization

Performance & Scaling: Redis, Caching, Rate Limiting, Load Testing, Clustering, NGINX

System Design: Multi-tenancy, RBAC, Session Management, High-Concurrency Systems

DevOps: Docker, Docker Compose, CI/CD, GitHub Actions

Frontend: React, Redux, Tailwind CSS

Languages & Tools: JavaScript, TypeScript, Git, Postman, Vercel, Render

EDUCATION

Bachelor of Computer Applications

Sarala Birla University, Ranchi

Sep 2023 – Present

CERTIFICATIONS

- Job-Ready AI-Powered Bootcamp – Sheryians Coding School
- Software Engineering Job Simulation – JPMorgan Chase & Co. (Forage)

LANGUAGES

- English (B2)
- Hindi (Native)